Submit your source code file as well as a brief report with the following components:

- for each epsilon value in {0.01, 0.1, 1, 10}, report the MAE for your differentially private histogram

0.01 - 85.82432432432432

0.1 - 9.763513513513514

1 - 0.8918918918918919

10 - 0.013513513513513514

- discuss the above observations and what you learn from this implementation exercise

Creating this program helped me better understand how noise was being added to data sets. Before starting I did not fully understand how noise was added and what variables affect noise. Now that I am done creating the program though I understand the method of how noise is added and the math that decides how much noise will be added based on your epsilon value.

- for each epsilon value in {0.01, 0.1, 1, 10}, run the differentially private histogram 10 times. For each run, compare the most frequent age-gender combination obtained using differentially private bin counts to the ground truth. Report the number of runs out of 10 where the most frequent age-gender combination is accurate, e.g., 0/10 or 7/10.

0.01 - 8/10

0.1 - 10/10

1 - 10/10

10 - 10/10